Internship Report:

Introduction

- **Project Title: Google Play Store Data Analytics Dashboard**
- **Domain: Data Analytics & Business Intelligence**
- **Objective: Analyze Google play store application using the performance metrics using the Google Play Store dataset(Play Store Data.csv And User Reviews.csv.**
- **Background and Context**

During my internship, I worked on processing and visualizing Dashboard according to the requirement of task which help user the analyze application data from the Google Play Store. The primary goal was to transform raw Google data into actionable insights for stakeholders looking to understand top-performing application categories.

Task1:Create Grouped bar Chart.

Specific Task Scope

The task required building a dynamic grouped bar chart to compare the average rating and total review count of the top 10 app categories ranked by the number of installs.

To ensure high data quality and relevance, the dataset was filtered using strict business logic:

- **Category Filter: Top 10 app categories by total number of installs.**
- **Rating Filter: Exclude any category with an average rating below `4.0`.**
- **Size Filter: Exclude any app size below `10 MB`.**
- **Regency Filter: Include only apps last updated in the month of January.**
- **Time-Based Access Control: Implement a security/visibility rule ensuring the graph renders on the dashboard exclusively between 3:00 PM IST and 5:00 PM IST, remaining hidden at all other times to support time-sensitive monitoring.**

Task2:An interactive Choropleth map using Plotly to visualize global installs by Category.

Specific Task Scope: For An interactive Choropleth map locations(Places) are required like region/State/Country . Locations are missing in the given dataset of "Google Play Store",so only data filtration's are done according to the requirement of the Task2.

To ensure high data quality and relevance, the dataset was filtered using strict business logic:

- **Filter1: Display the top 5 categories by highlighting regions/categories with > 1,000,000 installs.**
- **Filter2: Drop any category starting with the letters A, C, G, or S (case-insensitive).**
- **Visibility: Render exclusively between 18:00 and 20:00 IST.**

Task3:Create a dual axis bar graph comparing the average installs and revenue for the Free App vs Paid App within the Top 3 categories.

Target Scope: Top 3 app categories based on dataset volume or relevance.

- **Time Restriction: Visible and active only between 1:00 PM IST and 2:00 PM IST daily.**
- **To ensure high data quality and relevance, the dataset was filtered using strict business logic:**
- **APP Installs must be grater than 10,000.**
- **App Revenue: Greater than or equal to the 10,000.**
- **Android Version : Must be greater than or equal to the 4.0 version.**
- **App Size: Must be greater than 15MB.**
- **Content Rating: Must be for Everyone**
- **App Name Length: Must be grater than 30 characters including the spaces and special characters**
- **Visibility: Render exclusively between 13:00 and 14:00 IST.**

- **TASK 4:**
  Plot time Series line chart graph to show trend of total installs over time.

- **Objective:**
  Create a dashboard time-series line chart tracking monthly total installs over time, segmented by app category, with highlighted growth period.

- **Key Visuals: Shaded areas under the curve during periods where month-over-month (MoM) install growth exceeds 20%.**
- **Visibility: Active and visible exclusively between 6:00 PM IST and 9:00 PM IST daily.**

Data Filters Applied:

- **App Name Starting Letters: Cannot start with X, Y, or Z (case-insensitive).**

- **App Name Character Restriction: Cannot contain the letter "S" (case-insensitive) anywhere in the name.**

- **App Category Starting Letters:**

- **Must start with E, C, or B (e.g., Business, Communication, Education, Entertainment).**

- **Minimum Reviews: Must have ≥ 500 reviews.**

TASK 5:

Objective: A bubble chart to analyze the relationship between app size (in MB) and average rating, with the bubble size representing the number of installs.

- **Bubble Encoding: X-Axis maps App Size (MB), Y-Axis maps Average Rating, and the Bubble Size represents the total volume of Installs.**

Visibility: Visible and functional on the dashboard exclusively between 5:00 PM IST and 7:00 PM IST daily.

Data Filters Applied:

- **Target Categories: Game, Beauty, Business, Comics, Communication, Dating, Entertainment, Social, and Events.**
- **Minimum Rating: Greater than 3.5**
- **Minimum Reviews: Greater than 500.**
- **Minimum Installs: Greater than 50,000.**
- **Sentiment Subjectivity: Greater than 0.5 (Derived from user review sentiment analysis)**
- **App Name Character Restriction: Cannot contain the letter "S" or "s" anywhere in the string.**

- **Game Category: Must be styled with a distinctive Pink color highlight to separate it visually from the other categories**

- **Task6**:
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- **A stacked area chart to visualize the cumulative number of installs over time for each app category, with each category represented as a separate color band in the chart.**
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  Data Filters Applied:

- **1 Rating Filter: Include only Apps where Rating greater than equal to 4.2.**
  2.Application name doesn’t numbers 0- 9.
  3.Application Categories starting with 'T' or 'P'.
  4. Include only Apps where Reviews > 1,000 .
  5.Application Size must be between 20 MB and 80 MB

  Visibility Of Graph:

  Active Window: 04:00 PM IST – 06:00 PM IST (16:00 – 18:00 IST)

  Stacked Area Chart Specifications:

  X-Axis: Continuous timeline broken down into monthly intervals.

  Y-Axis: Cumulative summation of installs grouped chronologically.

  Visual Structure: Part-to-whole area blocks stacked sequentially, showing individual category volume shifts without layer overlap.

  Translate the Beauty category in Hindi and Business category in Tamil and Dating category in German to show on Graphs.